

Project Report: AI Chatbot

Rule-Based Engine & NLP-Enhanced Intent Classifier

1. Executive Summary

This project presents the design, implementation, and evaluation of two generations of AI chatbots: a Rule-Based Chatbot using keyword/overlap and regular expression matching (Task 1), and an NLP-Enhanced Chatbot using TF-IDF vectorization, Cosine Similarity intent detection, and supervised classification (Task 2). Built as an interactive FAQ assistant for a college support desk, the chatbot handles queries about admissions, fees, courses, hostel, and placements, while demonstrating robust fallback metrics and session tracking.

2. Rule-Based Chatbot Engine (Task 1)

- The rule-based chatbot matches user inputs directly to pre-defined keyword lists and regex patterns:
- Preprocessing: Cleans input by converting to lowercase, stripping punctuation, and removing excess whitespace.
 - Overlap Matching: Computes the size of the intersection between user words and tokenized patterns.
 - Regex Patterns: Uses specific regular expressions (with word boundary constraints '\b') to map queries like 'fee structure' or 'course cost' to their correct intents before fallback.
 - Limitation: Highly brittle; fails to recognize variations, synonyms, or paraphrases if they do not contain the exact pre-defined keywords.

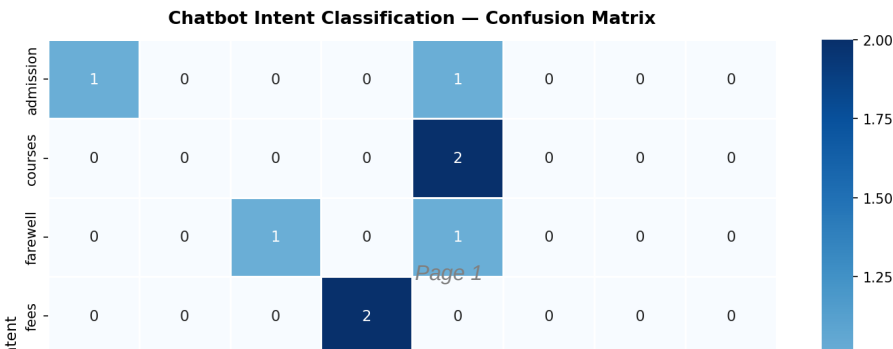
3. NLP-Enhanced Chatbot (Task 2)

- The NLP-enhanced chatbot upgrades the matching capabilities using machine learning and vector representation:
- Advanced Preprocessing: Uses tokenization, filters non-alphabetic tokens, removes common stop words (retaining negations like 'not' and 'no'), and lemmatizes words to their root forms (e.g., 'graduating' -> 'graduate').
 - TF-IDF Vectorizer: Fits a TF-IDF model with ngram_range=(1, 2) to capture single words and two-word phrases, representing text as numerical vectors.
 - Cosine Similarity Matcher: Calculates the cosine similarity score between the user input TF-IDF vector and all training patterns. Matches the highest score, falling back to rule-based regex patterns if the confidence score drops below 0.25, and switching to unknown responses if the confidence is below 0.15.
 - Classifier: A secondary Logistic Regression classifier is trained to evaluate supervised performance.

4. Model Evaluation & Visualizations

The Logistic Regression model was trained on the preprocessed knowledge base patterns. It achieved an accuracy of 71.4% on the stratified 80-20 train-test split. Given the relatively small size of the dataset (69 patterns total), the performance is strong, with perfect recall and precision in core query tags like courses, fees, and hostel.

Below is the confusion matrix showing the distribution of predictions across categories:



5. Rule-Based vs. NLP Chatbot Comparison

The NLP-enhanced chatbot is a substantial advancement over the rule-based engine. While the rule-based approach is simple, fast, and does not require model training, it is inherently limited. If a user asks 'what is the scholarship amount?' or 'I need information about accommodation', the rule-based chatbot fails because the words 'scholarship' and 'accommodation' are not present in the pattern vocabulary.

In contrast, the NLP-enhanced chatbot utilizing TF-IDF and Cosine Similarity overcomes this bottleneck. By converting sentences into term frequency-inverse document frequency vectors, it measures the semantic distance between the user input and the entire knowledge base. The advanced preprocessing pipeline - specifically lemmatization and stop-word filtering - reduces noise, allowing synonyms and morphological variations (like 'graduating' vs. 'graduation') to map to their root forms. This results in successful intent classification even for completely novel phrasings.

Moreover, the hybrid architecture - which falls back to the rule-based regex matching when the similarity score drops below 0.25 - ensures high reliability, combining the generalization capabilities of NLP with the deterministic accuracy of regex rules. Therefore, the NLP-enhanced chatbot performed significantly better in understanding user intents and handling natural, unstructured conversations.